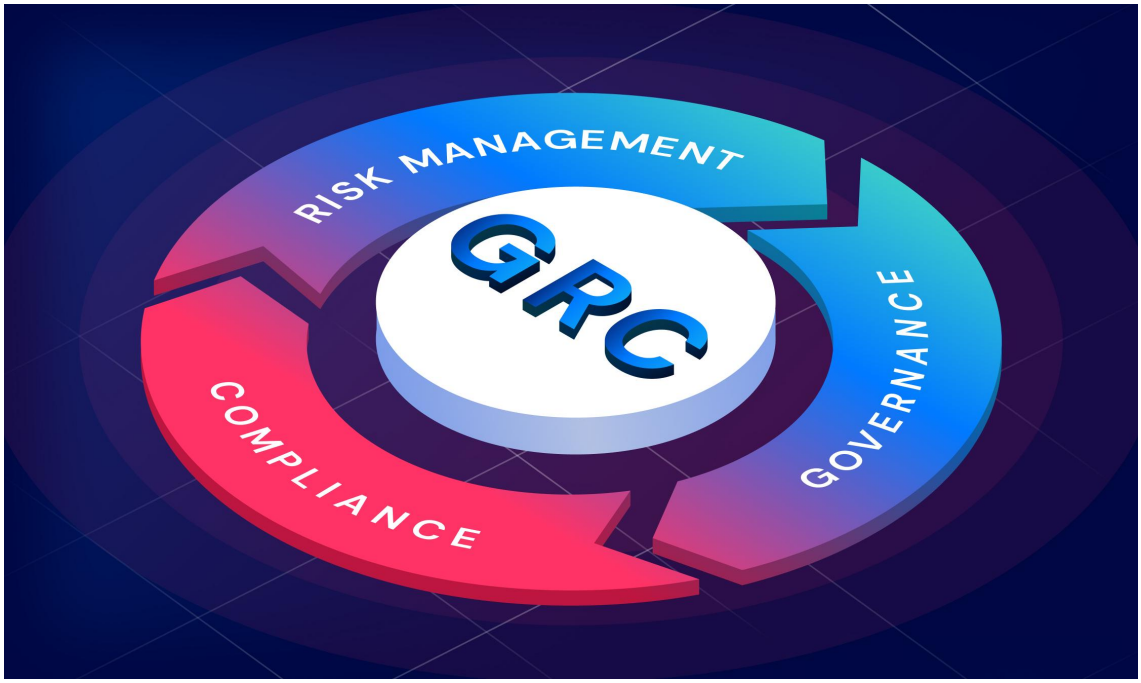
LINUX FUNDAMENTAL ASSIGNMENT



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Deliverables Step 1

- Username = output of whoami
- Hostname = output of hostname
- Current directory = output of pwd

Deliverables Step 2

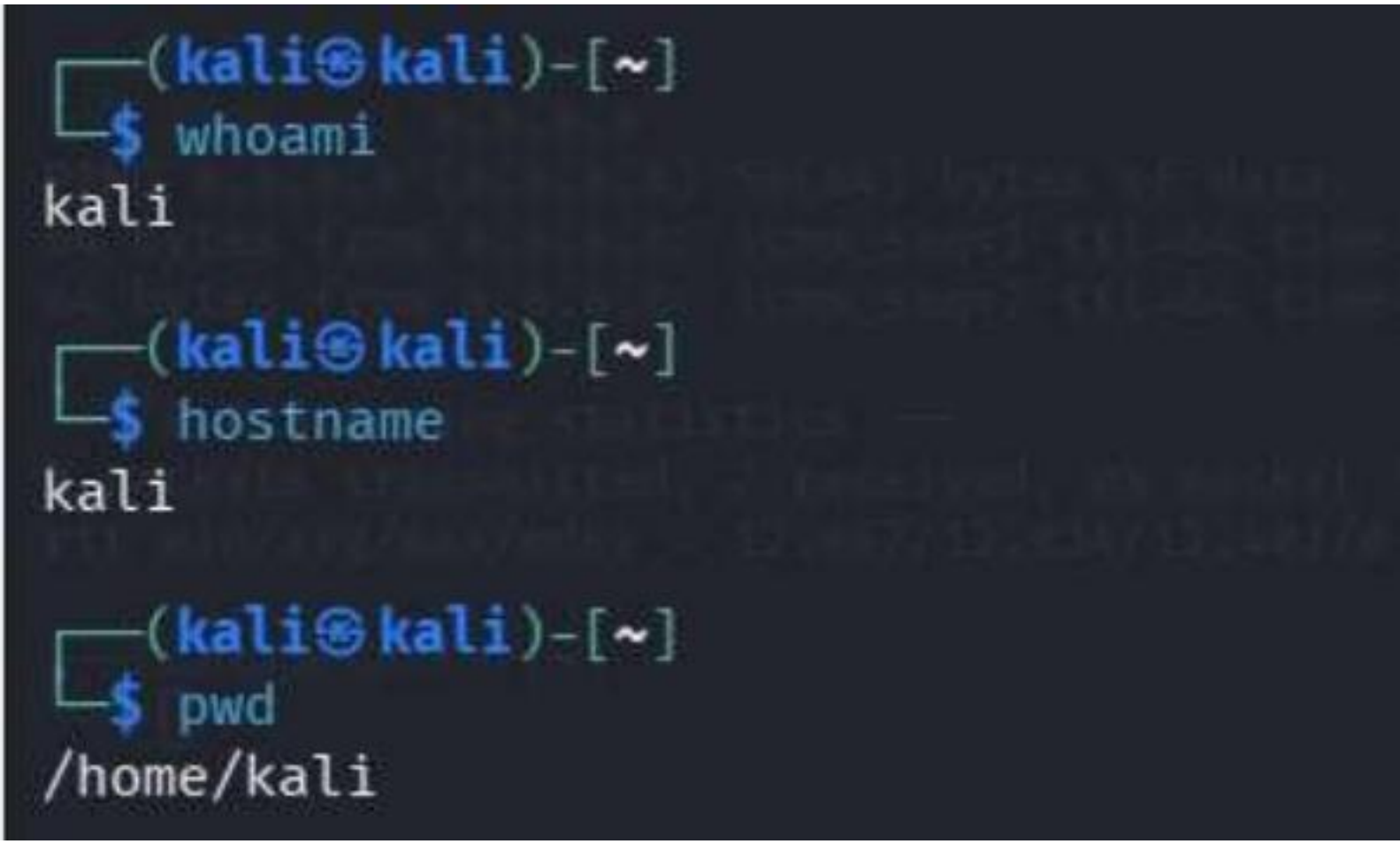
- Total RAM of the VM = 1.9Gi
- Available disk space on the main partition = 59G
- Kernel version = 6.18.12+kali=amd64

Deliverables Step 1

- The average round-trip time (avg RTT) = 13.034ms
- Packet loss = 0% (From the screenshot, 2 packets were sent, 2 packets were received successfully and no packets were lost during transmission.

Deliverables Step 2

- IP Address google.com resolves to = PING google.com (192.178.192.139). The round-trip time was similar to pinging 8.8.8.8 because both are Google servers and the network connection was stable. Slight differences in response time may occur due to routing and server location differences
- -c 4 means = Send only 4 ping packets and then stop automatically


(kali㉿kali)-[~]
$ whoami
kali
(kali㉿kali)-[~]
$ hostname
kali
(kali㉿kali)-[~]
$ pwd
/home/kali

```
┌──(kali㉿kali)-[~]
└─$ uname -a
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux
```

```
┌──(kali㉿kali)-[~]
└─$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       959Mi       340Mi        16Mi       829Mi       1.0Gi
Swap:          953Mi          0B       953Mi

┌──(kali㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            883M     0  883M   0% /dev
tmpfs           198M  968K  197M   1% /run
/dev/sda1        79G   16G   59G  21% /
tmpfs           986M  4.0K  986M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           986M   52K  986M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  108K  198M   1% /run/user/1000
```

```
┌──(kali㉿kali)-[~]
└─$ lscpu
Architecture:                       x86_64
  CPU op-mode(s):                   32-bit, 64-bit
  Address sizes:                    39 bits physical, 48 bits virtual
  Byte Order:                       Little Endian
CPU(s):                             2
  On-line CPU(s) list:              0,1
Vendor ID:                          GenuineIntel
  Model name:                       Intel(R) Core(TM) i5-8365U CPU @ 1.60GHz
    CPU family:                     6
    Model:                          142
    Thread(s) per core:             1
    Core(s) per socket:             2
    Socket(s):                      1
    Stepping:                       12
    BogoMIPS:                       3792.00
    Flags:                          fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall n
                                    x rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16
                                     pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch fsgsbas
                                    e bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_clear flush_l1d arch_capabilities
Virtualization features:
  Hypervisor vendor:                KVM
  Virtualization type:              full
Caches (sum of all):
  L1d:                              64 KiB (2 instances)
  L1i:                              64 KiB (2 instances)
  L2:                               512 KiB (2 instances)
  L3:                               12 MiB (2 instances)
NUMA:
  NUMA node(s):                     1
  NUMA node0 CPU(s):                0,1
Vulnerabilities:
  Gather data sampling:             Unknown: Dependent on hypervisor status
  Ghostwrite:                       Not affected
  Indirect target selection:        Mitigation; Aligned branch/return thunks
```

```
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit:             KVM: Mitigation: VMX unsupported
L1tf:                      Not affected
Mds:                       Not affected
Meltdown:                  Not affected
Mmio stale data:           Mitigation; Clear CPU buffers; SMT Host state unknown
Old microcode:             Not affected
Reg file data sampling:    Not affected
Retbleed:                  Vulnerable
Spec rstack overflow:      Not affected
Spec store bypass:         Vulnerable
Spectre v1:                Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI SW loop, KVM SW loop
Srbds:                     Unknown: Dependent on hypervisor status
Tsa:                       Not affected
Tsx async abort:           Mitigation; TSX disabled
Vmscape:                   Not affected
```


(kali㉿kali)-[~]
$ uptime
10:16:32 up 24 min, 1 user, load average: 0.23, 0.18, 0.20

```
┌──(kali㉿kali)-[~]
└─$ ping -c2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=13.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=12.7 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 12.667/13.034/13.401/0.367 ms
```

```
┌──(kali㉿kali)-[~]
└─$ ping -c 4 google.com
PING google.com (192.178.192.139) 56(84) bytes of data.
64 bytes from lcyyzn-in-f139.1e100.net (192.178.192.139): icmp_seq=1 ttl=64 time=12.0 ms
64 bytes from lcyyzn-in-f139.1e100.net (192.178.192.139): icmp_seq=2 ttl=64 time=12.5 ms
64 bytes from lcyyzn-in-f139.1e100.net (192.178.192.139): icmp_seq=3 ttl=64 time=12.3 ms
64 bytes from lcyyzn-in-f139.1e100.net (192.178.192.139): icmp_seq=4 ttl=64 time=9.99 ms

── google.com ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 9.993/11.706/12.532/1.011 ms
```